Rojae Mighty

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Research interests: High-energy nuclear physics, QCD and gluon saturation at the Electron-Ion Collider, computational data analysis, and machine learning for physical and financial time-series systems.

Education

Stony Brook University *Expected May 2027*
B.S. in Physics
Relevant coursework: quantum mechanics, electrodynamics, statistical mechanics, computational methods, data analysis

Research Experience

Brookhaven National Laboratory *2025–Present*

- **Mini-Semester Program** (2025) — intensive introduction to nuclear science at BNL; first research exposure with STAR/EIC physics.
- **DOE Science Undergraduate Laboratory Internship (SULI)** (Summers 2025, 2026) — electron-ion collision studies of semi-inclusive processes; implemented a spatially dependent parton distribution function using **ROOT** and **Python**.

Center for Frontiers in Nuclear Science (CFNS) *2025–Present*
Undergraduate Researcher — Nuclear & High-Energy Physics
Advisor: Zhoudunming (Kong) Tu

- Study gluon saturation and geometric scaling for the Electron-Ion Collider (EIC) to probe QCD dynamics at small x.
- Develop machine-learning methods for Deep Inelastic Scattering kinematic reconstruction (extracting x and Q^2) for inclusive and extended scattering channels.
- Build analysis workflows in **C++**, **ROOT**, and **Python**; run numerical simulations and Monte Carlo-based studies on high-energy collision data.

Honors & Awards

- Barry Barish Undergraduate Research Travel Award, Stony Brook University (2026)
- Edward Guiliano Global Fellowship — *Myth and Matter: Exploring Early Explanations of Nature*, Greece (2026)
- 1st Place, Physical Sciences Presentation, Yale Undergraduate Research Conference (2026)
- Outstanding Oral Presentation Award, National Society of Black Physicists Conference (2025)

Presentations & Conferences

Geometric Scaling and Gluon Saturation at the EIC

- Deep Inelastic Scattering Conference (DIS 2026), Bologna, Italy — oral presentation
- Yale Undergraduate Research Conference (2026) — poster presentation
- National Society of Black Physicists Annual Conference (2025) — oral presentation
- APS DNP Meeting (2025, CEU) — poster presentation
- APS Global Physics Summit (2026) — oral presentation
- Stony Brook Physics Colloquium (2026) — invited oral presentation

Projects

Market Microstructure Forecasting (GitHub) *2025–Present*

Independent — Early-stage quantitative ML project

- Building a Python ML pipeline on limit order book data to forecast short-horizon mid-price returns and volatility.
- Engineering microstructure features (order-flow imbalance, spread dynamics); implementing walk-forward validation and transaction-cost-aware backtesting.

Leadership & Service

Dean's Student Leadership & Advisory Council (DSLAC), College of Arts & Sciences *2024–Present*

- Represent undergraduates to the CAS Dean on policy and campus services; CAS Ambassador at admissions events and Godfrey Excellence in Teaching Awards.

Collegiate Science and Technology Entry Program (CSTEP), Stony Brook University *2024–Present*

- Peer mentor for underrepresented STEM undergraduates on research pathways, academic planning, and professional development.

Socially Just Seawolves (SJS) Living-Learning Community *2023–Present*

- Lead food-insecurity awareness programming with campus support services; service-learning and community agriculture volunteering.

Technical Skills

Programming:	Python, C++, Unix/Linux shell scripting
Research tools:	ROOT, LaTeX, Git, Monte Carlo event generators, numerical simulation
Methods:	Machine learning (scikit-learn), statistical data analysis, walk-forward validation, feature engineering
Languages:	English (native), French (elementary)